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Refrigeration and Heat Pump Technology

Final Project

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1 Introduction

This project analyzes a vapor-compression refrigeration system for data center cooling with a nominal capacity of 50 kW. The objective is to evaluate the variation of cooling capacity (Q_e) and Energy Efficiency Ratio (EER) with condenser heat sink temperature.

A parametric study is performed by varying the condenser water inlet temperature while keeping all other parameters fixed.

The implementation is developed as a **modular Python codebase** hosted on GitHub. While the primary objective is the analysis of the author's specific assignment, the code is structured so that any student from the course can run their own project by editing two configuration files. This is documented in the `universal-model` branch of the repository.

1.1 Given Data

Table 1: Project specifications provided in the assignment

Parameter	Value
Application	Data center cooling
Nominal cooling capacity	50 kW
Evaporator condition	Air at 12 °C
Condenser secondary fluid	Water

1.2 System and Modeling Overview

The system is modeled as a single-stage vapor-compression cycle using R32. A **Bitzer GSU60182VL_4 (ORBIT+)** compressor is selected.

Compressor performance is evaluated using manufacturer-based polynomial correlations, while thermodynamic properties are computed with **CoolProp**. Heat exchangers are modeled using the LMTD method, and operating temperatures are obtained iteratively.

Additional modeling assumptions and design choices are detailed in the following sections.

2 System Definition and Design Choices

2.1 Application Requirements

The system is designed for data center cooling with a nominal capacity of 50 kW. Air is cooled from 12 °C, while heat is rejected to a water-based heat sink with a temperature rise of 5 K.

A parametric study is performed by varying the condenser water inlet temperature between 20 °C and 40 °C to evaluate its impact on cooling capacity and EER.

2.2 Refrigerant Selection

The selected refrigerant is **R32**. The choice is based on thermodynamic performance, compatibility with compressor data, and suitability for medium-temperature cooling applications.

Advantages:

- High volumetric cooling capacity → compact system design
- Good heat transfer properties → improved efficiency
- Lower GWP compared to traditional refrigerants (e.g., R134a)

Limitations:

- Mild flammability (A2L classification)
- Higher discharge temperatures

Despite these limitations, R32 is well suited for this application due to its high efficiency and widespread use in modern HVAC systems. In addition, the availability of **manufacturer data (Bitzer)** for R32 enables accurate compressor modeling.

Thermodynamic properties are evaluated using **CoolProp**.

2.3 Compressor Selection

A **scroll compressor (Bitzer ORBIT+)** is selected. Compared to reciprocating, screw, or centrifugal compressors, scroll compressors are more suitable for this application due to:

- High efficiency at small-to-medium capacities (50 kW)
- Low vibration and compact design
- Good performance under variable operating conditions

Alternative compressor types were not selected because: - Reciprocating: higher mechanical losses and maintenance - Screw: more suitable for larger capacities - Centrifugal: inefficient at this scale

The selected model is:

Table 2: Selected compressor specifications

Parameter	Value
Model	GSU60182VL_4
Series	ORBIT+
Type	Scroll compressor
Frequency	50 Hz
Capacity control	None
Swept volume	30.2 m^3/h

Nominal Operating Point

The manufacturer nominal point is used for validation:

Table 3: Nominal operating conditions (Bitzer)

Parameter	Value
T_{evap}	5.0 °C
T_{cond}	35.0 °C
Cooling capacity	53.9 kW
Compressor power	9.31 kW
Mass flow rate	735 kg/h
Discharge temperature	73.3 °C
COP	5.79

This point is used to verify consistency between the polynomial model and thermodynamic reconstruction.

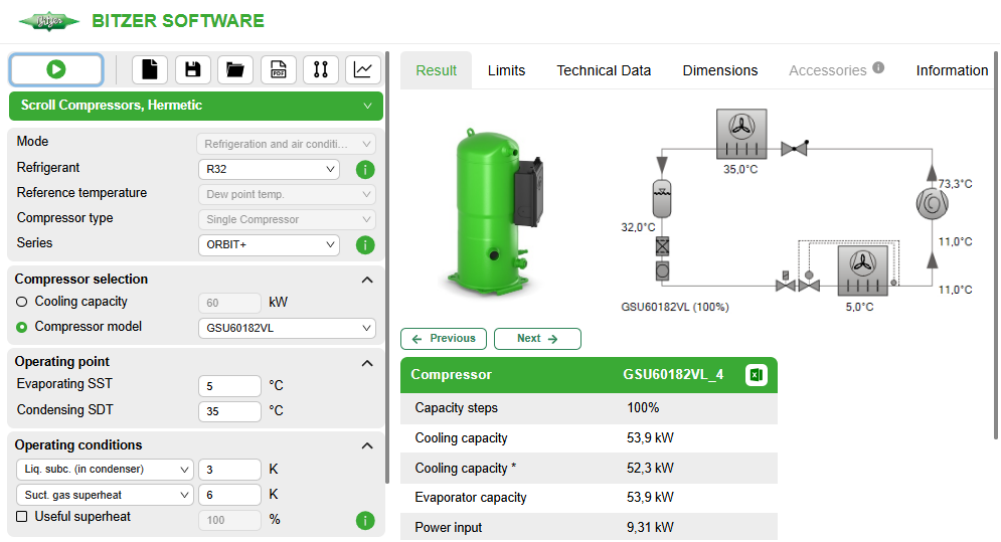

Figure 1: Selected compressor and nominal operating conditions from Bitzer software

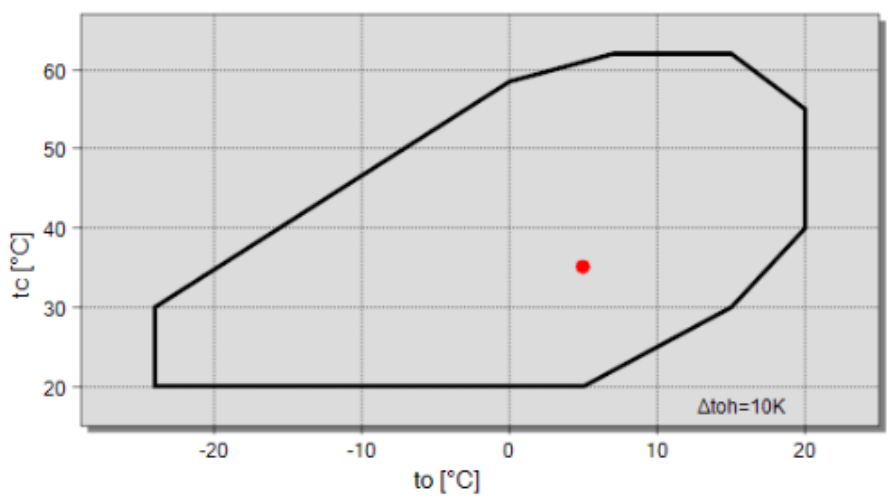

Figure 2: Compressor operating envelope (Bitzer) with selected nominal operating point.

The selected compressor operating conditions are verified against the manufacturer operating envelope shown in Figure 2. The nominal operating point lies well within the permissible region, ensuring that the compressor operates within its design limits. This confirms the compatibility of the chosen evaporating and condensing temperatures with the compressor constraints, including superheating requirements.

2.4 Heat Exchanger Modeling

Both evaporator and condenser are modeled as counterflow heat exchangers using the LMTD method with constant KA values determined at nominal conditions.

The operating temperatures are obtained iteratively from:

$$Q = KA \cdot \Delta T_{\text{lm}} \tag{1}$$

Evaporator:

$$Q_e = KA_{\text{evap}} \cdot \Delta T_{\text{lm,evap}} \tag{2}$$

Condenser:

$$Q_c = KA_{\text{cond}} \cdot \Delta T_{\text{lm,cond}} \tag{3}$$

This formulation ensures consistency between heat exchangers and cycle performance.

3 Thermodynamic Modeling of the Cycle

3.1 Cycle Description

The system is modeled as a single-stage vapor-compression cycle using R32. The four processes are:

1. Compression ($1 \to 2$)
2. Heat rejection ($2 \to 3$)
3. Expansion ($3 \to 4$)
4. Heat absorption ($4 \to 1$)

Thermodynamic properties are evaluated using **CoolProp**.

3.2 Model Assumptions

Table 4: Main modeling assumptions

Assumption	**Value / Description**
Operation	Steady-state
Pressure drops	Negligible
Superheating (ΔT_{sh})	6 K
Subcooling (ΔT_{sc})	3 K
Expansion process	Isenthalpic ($h_4 = h_3$)
Heat exchangers	Counterflow, constant KA

3.3 Thermodynamic States

The main states are defined as:

$$T_1 = T_{\text{evap}} + \Delta T_{sh} \tag{4}$$

$$h_2 = h_1 + \frac{P_c}{\dot{m}} \tag{5}$$

$$T_3 = T_{\text{cond}} - \Delta T_{sc} \tag{6}$$

$$h_4 = h_3 \tag{7}$$

All properties (h, s, T, P) are obtained using **CoolProp**.

3.4 Performance Parameters

$$Q_e = \dot{m}(h_1 - h_4) \tag{8}$$

$$P_c = \dot{m}(h_2 - h_1) \tag{9}$$

$$\text{EER} = \frac{Q_e}{P_c} \tag{10}$$

4 Compressor Modeling

4.1 Modeling Approach

Ideal compressor models based on constant efficiencies or pressure ratio correlations do not capture the strong dependence of performance on operating conditions.

For this reason, a **manufacturer-based polynomial model (Bitzer)** is adopted. This approach provides cooling capacity (Q_e), compressor power (P_c), and mass flow rate ($\dot{m}$) as continuous functions of evaporating and condensing temperatures.

4.2 Polynomial Model

The compressor performance is expressed using a 10-coefficient polynomial:

$$\begin{aligned} y = c_1 + c_2 T_{\text{evap}} + c_3 T_{\text{cond}} + c_4 T_{\text{evap}}^2 + c_5 T_{\text{evap}} T_{\text{cond}} + c_6 T_{\text{cond}}^2 \\ + c_7 T_{\text{evap}}^3 + c_8 T_{\text{cond}} T_{\text{evap}}^2 + c_9 T_{\text{evap}} T_{\text{cond}}^2 + c_{10} T_{\text{cond}}^3 \end{aligned} \tag{11}$$

where $y \in \{Q_e, P_c, \dot{m}\}$ and temperatures are expressed in °C.

Three independent coefficient sets are used for each variable. The model is evaluated at each operating condition to obtain continuous compressor behavior.

In the modular implementation, all coefficient sets are stored in a `COMPRESSOR_REGISTRY` dictionary in `src/compressor.py`. Each entry corresponds to a specific compressor model and refrigerant. The active compressor is selected via the `COMPRESSOR_MODEL` key in `src/config.py`, without modifying any computational logic.

4.3 Compressor Efficiencies

The polynomial model directly provides Q_e, P_c, and $\dot{m}$. Compressor efficiencies are derived using thermodynamic properties from **CoolProp**.

Isentropic Efficiency

$$\eta_{\text{is}} = \frac{h_{2s} - h_1}{h_2 - h_1} \tag{12}$$

with:

$$h_2 - h_1 = \frac{P_c}{\dot{m}} \tag{13}$$

The isentropic enthalpy h_{2s} is obtained from CoolProp at constant entropy.

Volumetric Efficiency

$$\eta_{\text{vol}} = \frac{\dot{m}}{\rho_1 \dot{V}_{\text{swept}}} \tag{14}$$

where ρ_1 is the suction density and $\dot{V}_{\text{swept}}$ is the compressor swept volume.

4.4 Model Consistency

At the nominal operating conditions ($T_{\text{evap}} = 5$°C, $T_{\text{cond}} = 35$°C), the polynomial model reproduces the manufacturer data for cooling capacity and power input with high accuracy, confirming its reliability within the considered operating range.

5 Heat Exchanger Modeling

5.1 LMTD Method

Heat transfer in both evaporator and condenser is modeled using the LMTD method:

$$Q = KA \cdot \Delta T_{\text{lm}} \tag{15}$$

For counterflow heat exchangers:

$$\Delta T_{\text{lm}} = \frac{\Delta T_1 - \Delta T_2}{\ln\left(\frac{\Delta T_1}{\Delta T_2}\right)} \tag{16}$$

For $\Delta T_1 \approx \Delta T_2$, the approximation is used:

$$\Delta T_{\text{lm}} \approx \frac{\Delta T_1 + \Delta T_2}{2} \tag{17}$$

5.2 Heat Exchanger Parameters

The overall conductances (KA) are assumed constant and determined at nominal conditions:

$$KA_{\text{evap}} = \frac{Q_e}{\Delta T_{\text{lm,evap}}} \tag{18}$$

$$KA_{\text{cond}} = \frac{Q_c}{\Delta T_{\text{lm,cond}}} \tag{19}$$

Table 5: Heat exchanger conductances at nominal conditions

Parameter	Value
KA_{evap}	11.43 kW/K
KA_{cond}	8.77 kW/K

These values are kept constant throughout the analysis.

5.3 Evaporator Model

$$\Delta T_1 = T_{\text{air,in}} - T_{\text{evap}}, \quad \Delta T_2 = T_{\text{air,out}} - T_{\text{evap}} \tag{20}$$

$$Q_e = KA_{\text{evap}} \cdot \Delta T_{\text{lm,evap}} \tag{21}$$

The evaporating temperature T_{evap} is obtained by solving the nonlinear heat balance. In the general implementation, this solver accepts any secondary fluid (air, water, or brine) via the `EVAP_SECONDARY` parameter in the configuration.

5.4 Condenser Model

$$\Delta T_1 = T_{\text{cond}} - T_{\text{water,out}}, \quad \Delta T_2 = T_{\text{cond}} - T_{\text{water,in}} \tag{22}$$

$$Q_c = KA_{\text{cond}} \cdot \Delta T_{\text{lm,cond}} \tag{23}$$

The condensing temperature T_{cond} is obtained from the corresponding heat balance. Similarly, the condenser model supports both water-cooled and air-cooled configurations via the `COND_SECONDARY` parameter.

6 Numerical Solution Strategy

6.1 Coupled Problem

The evaporating and condensing temperatures are determined by coupling the thermodynamic cycle with the heat exchanger models. They must satisfy:

$$Q_e = KA_{\text{evap}} \cdot \Delta T_{\text{lm,evap}} \tag{24}$$

$$Q_c = KA_{\text{cond}} \cdot \Delta T_{\text{lm,cond}} \tag{25}$$

Since Q_e and Q_c depend on T_{evap} and T_{cond}, the problem is solved iteratively.

6.2 Iterative Procedure

For each operating condition:

1. Initialize:
$$T_{\text{evap}}^{(0)}, \quad T_{\text{cond}}^{(0)} \tag{26}$$
2. At iteration k, compute:
$$Q_e^{(k)},\ Q_c^{(k)},\ \dot{m}^{(k)},\ P_c^{(k)} \tag{27}$$
3. Update temperatures from heat exchanger balances:
$$T_{\text{evap,new}}^{(k)}, \quad T_{\text{cond,new}}^{(k)} \tag{28}$$
4. Apply relaxation:
$$T_{\text{evap}}^{(k+1)} = (1-\alpha)T_{\text{evap}}^{(k)} + \alpha T_{\text{evap,new}}^{(k)} \tag{29}$$
$$T_{\text{cond}}^{(k+1)} = (1-\alpha)T_{\text{cond}}^{(k)} + \alpha T_{\text{cond,new}}^{(k)} \tag{30}$$
5. Repeat until convergence.

A relaxation factor $\alpha = 0.45$ is used.

6.3 Convergence

Convergence is achieved when:

$$\max\left(\left|T_{\text{evap}}^{(k+1)} - T_{\text{evap}}^{(k)}\right|, \left|T_{\text{cond}}^{(k+1)} - T_{\text{cond}}^{(k)}\right|\right) < 10^{-4}\ \text{K} \tag{31}$$

A maximum of $k_{\max} = 100$ iterations is imposed. Convergence is typically reached within 17–18 iterations.

7 Parametric Study

7.1 Operating Conditions

The analysis is performed by varying the condenser water inlet temperature while keeping all other parameters constant.

Table 6: Fixed operating conditions

Parameter	Value
Refrigerant	R32
Cooling capacity target	50 kW
Evaporator air (in/out)	12 / 8 °C
Water temperature rise	5 K
Superheating	6 K
Subcooling	3 K
KA_{evap}	11.43 kW/K
KA_{cond}	8.77 kW/K

7.2 Variable Parameter

The condenser water inlet temperature is varied as:

$$T_{\text{water,in}} = 20^\circ\text{C to } 40^\circ\text{C} \tag{32}$$

with a step of 2°C, and:

$$T_{\text{water,out}} = T_{\text{water,in}} + 5\text{ K} \tag{33}$$

7.3 Simulation Outputs

For each operating condition, the following quantities are evaluated:

- Cooling capacity (Q_e)
- Compressor power (P_c)
- Condenser heat rejection (Q_c)
- Energy Efficiency Ratio (EER)
- Pressure ratio
- Isentropic and volumetric efficiencies

8 Results

8.1 Validation at Nominal Operating Point

The model is validated against the manufacturer nominal point:

Table 7: Nominal operating point comparison

Parameter	Manufacturer	Model
T_{evap} [°C]	5.0	5.0
T_{cond} [°C]	35.0	35.0
Q_e [kW]	53.9	53.97
P_c [kW]	9.31	9.31
$\dot{m}$ [kg/s]	0.2042	0.2041
COP / EER	5.79	5.796

Additional computed quantities:

- $\eta_{\text{is}} = 0.7484$
- $\eta_{\text{vol}} = 0.9793$

The agreement confirms the correctness of the implemented model.

8.2 Thermodynamic State Analysis

At $T_{\text{water,in}} = 30°\text{C}$:

$$T_{\text{evap}} = 5.16°\text{C}, \quad T_{\text{cond}} = 39.89°\text{C} \tag{34}$$

Table 8: Thermodynamic states

State	T [°C]	P [bar]	h [kJ/kg]	s [kJ/kg-K]
1	11.16	9.56	523.68	2.1625
2	81.20	24.72	574.09	2.1944
3	36.89	24.72	269.01	–
4	5.16	9.56	269.01	1.2478

The results confirm expected cycle behavior (superheating, subcooling, and isenthalpic expansion).

8.3 Performance Trends

System performance is evaluated for $T_{\text{water,in}} = 20°\text{C}$ to $40°\text{C}$.

Cooling Capacity

Q_e decreases from 55.89 kW to 47.75 kW (-14.6%) due to reduced evaporator enthalpy difference.

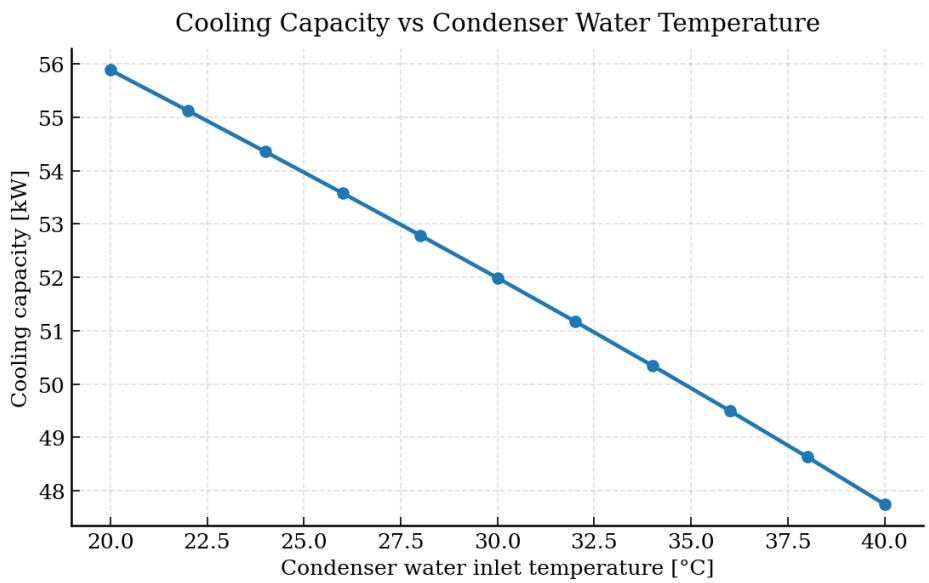

Figure 3: Cooling capacity vs condenser water inlet temperature

Compressor Power

P_c increases from 8.43 kW to 12.60 kW (+49.5%) due to higher compression work.

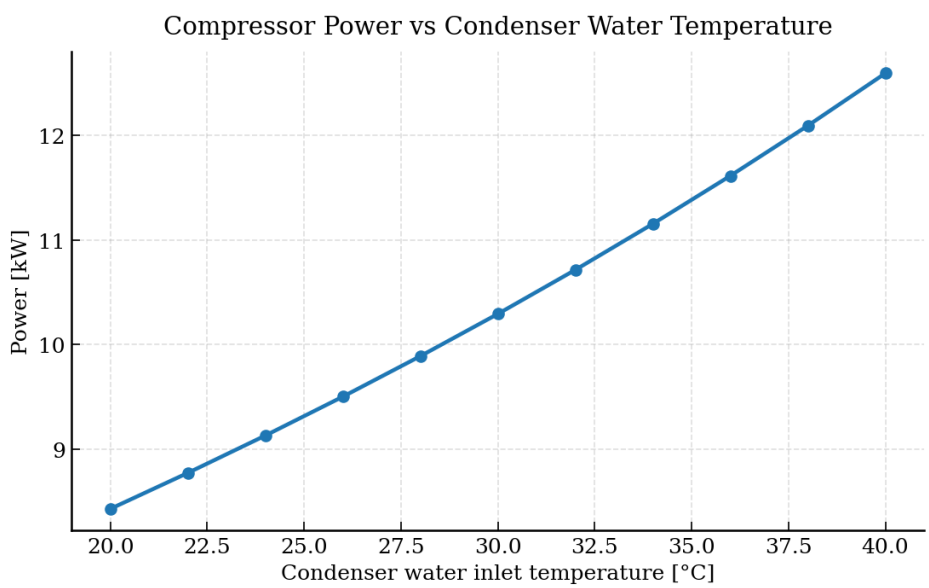

Figure 4: Compressor power vs condenser water inlet temperature

Energy Efficiency Ratio

EER decreases from 6.63 to 3.79 (−42.9%), indicating efficiency degradation at higher heat sink temperatures.

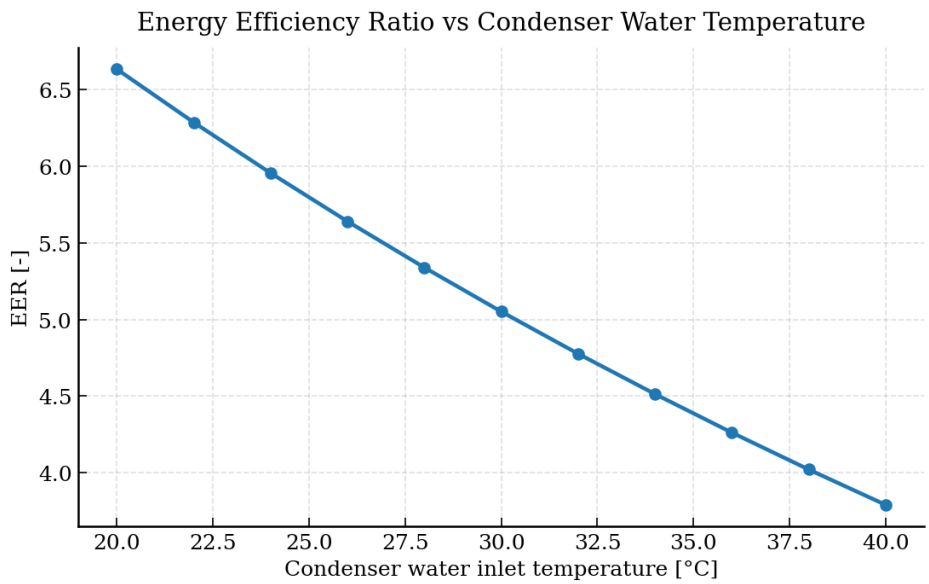

Figure 5: EER vs condenser water inlet temperature

Pressure Ratio

The pressure ratio increases from 2.04 to 3.23, driven by rising condensing pressure.

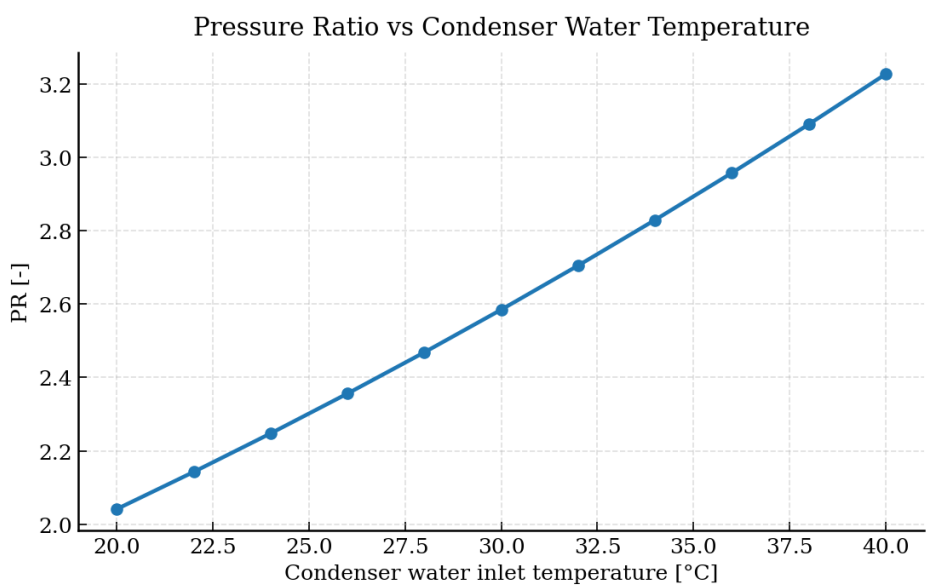

Figure 6: Pressure ratio vs condenser water inlet temperature

Compressor Efficiencies

η_{is} increases slightly (0.70 → 0.80), while η_{vol} decreases slightly (0.98 → 0.96).

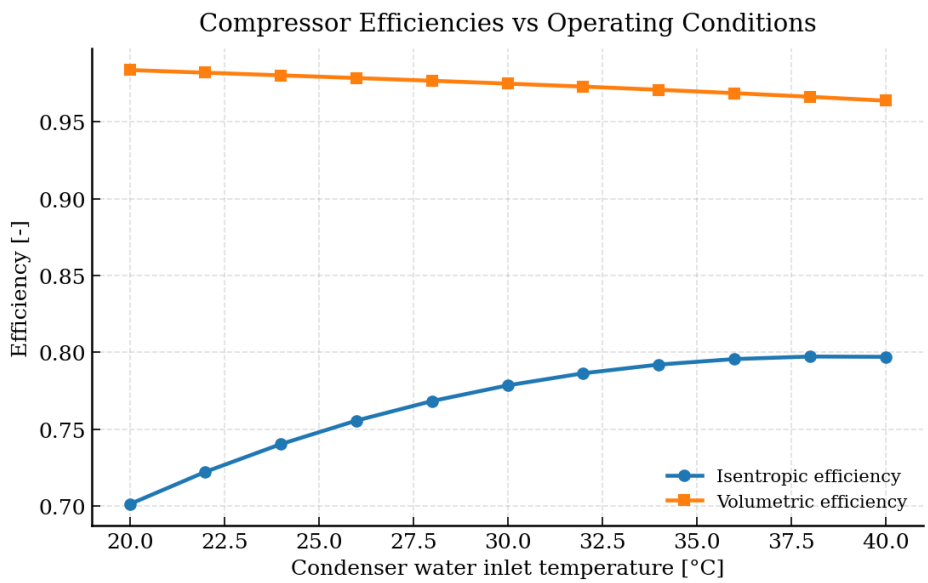

Figure 7: Compressor efficiencies vs condenser water inlet temperature

8.4 Model Consistency

Cooling capacity and compressor power obtained from the polynomial model are consistent across the operating range. This confirms the reliability of the compressor model and the overall numerical implementation.

9 Thermodynamic Diagrams

9.1 P-h Diagram

The pressure–enthalpy (P–h) diagram is used to visualize the thermodynamic cycle and the behavior of the refrigerant across the different components. The saturation dome is included to distinguish liquid and vapor regions, and the cycle is plotted for representative operating conditions.

The four main processes (compression, heat rejection, expansion, and heat absorption) are clearly identified. Within the heat exchangers, the processes are further separated into desuperheating, condensation, subcooling, evaporation, and superheating, allowing a more detailed interpretation of the cycle.

Multiple operating conditions (e.g., 24°C, 30°C, and 36°C condenser water inlet temperatures) are shown on the same diagram. As the condenser temperature increases, the cycle shifts toward higher pressure and enthalpy levels, leading to increased compression work.

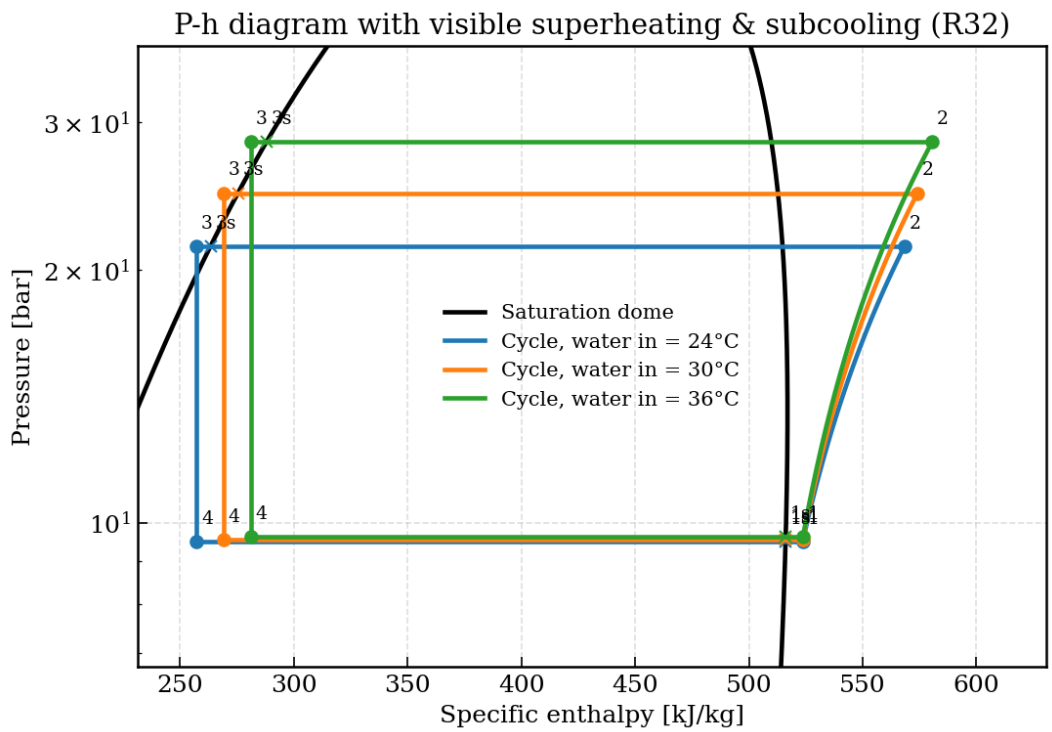

Figure 8: Pressure–enthalpy (P–h) diagram for selected operating conditions

9.2 T-s Diagram

The temperature–entropy (T–s) diagram provides a complementary representation of the cycle, highlighting thermal processes and entropy variations.

The saturation dome is plotted in the (T, s) plane, with the cycle superimposed for selected operating conditions. The compression process shows an increase in both temperature and entropy, while condensation and evaporation occur approximately at constant pressure. The expansion process is characterized by an increase in entropy due to irreversibilities.

This diagram also allows visualization of the deviation between the actual compression process and the ideal isentropic path.

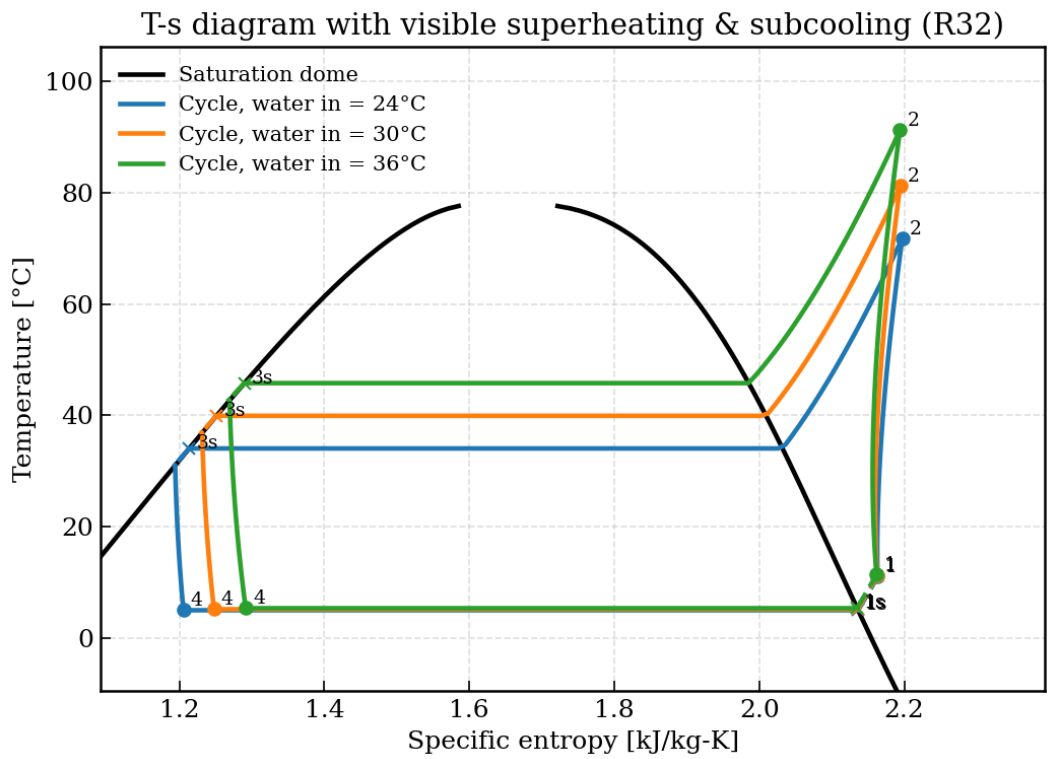

Figure 9: Temperature–entropy (T–s) diagram for selected operating conditions

9.3 Superheating and Subcooling Representation

Superheating and subcooling regions are explicitly highlighted in both diagrams.

In the P–h diagram, superheating extends beyond the saturated vapor line, while subcooling extends beyond the saturated liquid line. In the T–s diagram, these regions appear as shifts away from the saturation curves.

Dashed lines are used to distinguish these segments from the main phase-change processes, improving readability.

The imposed conditions:

$$\Delta T_{\text{sh}} = 6\ \text{K}, \quad \Delta T_{\text{sc}} = 3\ \text{K} \tag{35}$$

are clearly reflected in the diagrams, illustrating the deviation from the ideal saturated cycle and their impact on system behavior.

10 Extended Analysis

10.1 High Pressure Ratio Behavior

An extended analysis is performed to investigate compressor behavior at high pressure ratios. This is achieved by fixing $T_{\text{evap}} = 5°\text{C}$ and increasing the condensing temperature up to approximately 70°C.

The pressure ratio is defined as:

$$PR = \frac{P_{\text{cond}}}{P_{\text{evap}}} \tag{36}$$

As T_{cond} increases, PR rises significantly, leading to the following trends:

- η_{is} increases initially and then stabilizes,
- η_{vol} decreases progressively.

These behaviors are consistent with compressor physics: higher pressure ratios increase compression work and internal losses, while volumetric efficiency decreases due to re-expansion effects.

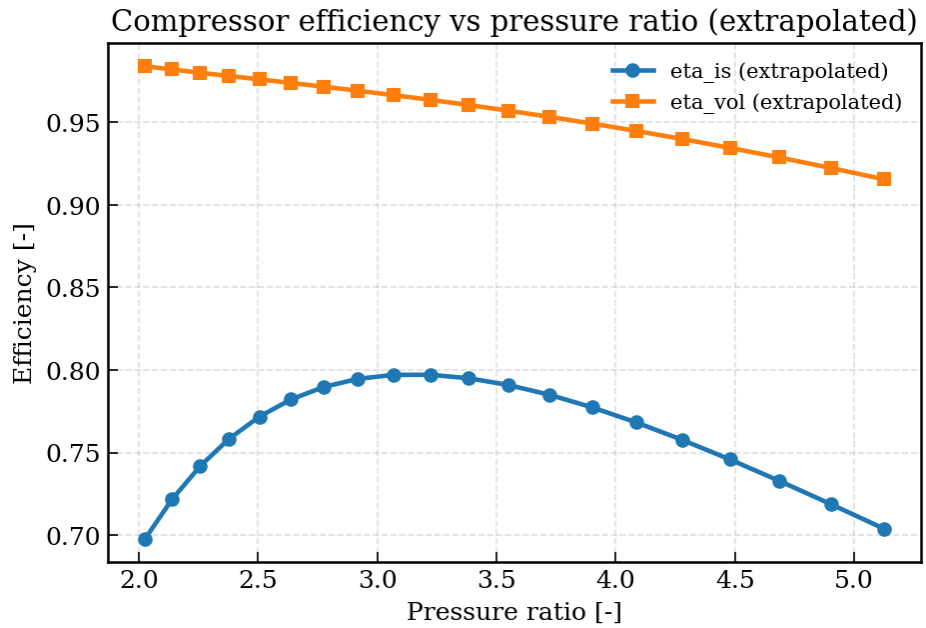

Figure 10: Compressor efficiencies as functions of pressure ratio (extended range)

10.2 Model Extrapolation

The polynomial compressor model provides continuous predictions and can be evaluated outside the original data range. However, extrapolated results must be interpreted with caution.

- No explicit constraints on mechanical or operational limits are included,
- The model does not distinguish between interpolation and extrapolation,
- Accuracy may decrease outside the calibrated operating region.

Therefore, this analysis is intended for qualitative insight only. The observed trends are useful for understanding compressor behavior, but should not be used for design or operation beyond the validated range.

11 Project Consistency with Requirements

11.1 Compliance with Methodology

The developed model strictly follows the methodology outlined in the project guidelines. The key required steps are all implemented and verified:

- **Compressor selection:** A specific compressor (Bitzer GSU60182VL_4) is selected based on manufacturer data, as required.
- **Use of polynomial model:** The compressor performance is modeled using a 10-coefficient polynomial formulation, allowing continuous evaluation of cooling capacity, power, and mass flow rate as functions of T_{evap} and T_{cond}.

- **Derivation of efficiencies:** The isentropic and volumetric efficiencies are not assumed but derived from the polynomial outputs combined with thermodynamic properties:

$$\eta_{\text{is}} = \frac{h_{2s} - h_1}{P_c/\dot{m}}, \quad \eta_{\text{vol}} = \frac{\dot{m}}{\rho_1 \dot{V}_{\text{swept}}} \tag{37}$$

- **Thermodynamic modeling:** The full vapor-compression cycle is modeled using CoolProp for accurate property evaluation.
- **Heat exchanger modeling:** The evaporator and condenser are modeled using the LMTD method with constant KA values, determined from nominal conditions.
- **Iterative coupling:** The evaporating and condensing temperatures are obtained through an iterative solution that ensures consistency between the cycle and the heat exchangers.
- **Parametric study:** The effect of varying condenser water inlet temperature on cooling capacity and EER is analyzed over a defined range.

These elements collectively ensure that the implementation is fully aligned with the required workflow, including the use of manufacturer-based data, thermodynamic consistency, and parametric analysis of system performance.

12 Conclusion

A thermodynamic model of a vapor-compression refrigeration system for data center cooling has been developed and analyzed. The study evaluated the variation of cooling capacity and Energy Efficiency Ratio (EER) as a function of the condenser heat sink temperature.

The model combines a manufacturer-based polynomial compressor representation with **CoolProp** thermodynamic calculations and LMTD-based heat exchanger models. The operating conditions are obtained through an iterative coupling, ensuring physical consistency of the cycle.

The results show that increasing the condenser water inlet temperature leads to:

- a decrease in cooling capacity (-14.6%),
- a significant decrease in EER (-42.9%),
- an increase in compressor power ($+49.5\%$).

These trends are driven by higher condensing temperatures and pressure ratios, resulting in increased compression work and reduced efficiency.

The analysis confirms that the polynomial compressor model provides a reliable representation of performance across the studied operating range. The extended analysis further highlights the sensitivity of compressor behavior to pressure ratio variations.

Beyond the primary project objectives, the implementation was developed with modularity and reusability as explicit design goals. The resulting codebase, described in detail in Appendix 12, is structured so that any student from the course can adapt it to their own project by editing two configuration files, without modifying the computational core.

APPENDIX

A. POLYNOMIAL COEFFICIENTS

The compressor performance is evaluated using three independent 10-coefficient polynomials corresponding to cooling capacity (Q_e), compressor power (P_c), and refrigerant mass flow rate ($\dot{m}$).

The general polynomial form is:

$$\begin{aligned} y = c_1 + c_2 T_{\text{evap}} + c_3 T_{\text{cond}} + c_4 T_{\text{evap}}^2 + c_5 T_{\text{evap}} T_{\text{cond}} + c_6 T_{\text{cond}}^2 \\ + c_7 T_{\text{evap}}^3 + c_8 T_{\text{cond}} T_{\text{evap}}^2 + c_9 T_{\text{evap}} T_{\text{cond}}^2 + c_{10} T_{\text{cond}}^3 \end{aligned} \tag{38}$$

The coefficients used in this work are:

Cooling Capacity (Q_e):

$c_1 = 59389.6461$ $c_2 = 1974.9334$ $c_3 = -434.5662$
$c_4 = 24.5945$ $c_5 = -10.2182$ $c_6 = 2.2480$
$c_7 = 0.1388$ $c_8 = -0.1284$ $c_9 = -0.02675$
$c_{10} = -0.03263$

Compressor Power (P_c):

$c_1 = 3944.2859$ $c_2 = 31.3520$ $c_3 = 132.7176$
$c_4 = 2.6430$ $c_5 = -0.1076$ $c_6 = -0.3117$
$c_7 = 0.05561$ $c_8 = -0.03643$ $c_9 = -0.001422$
$c_{10} = 0.02178$

Mass Flow Rate ($\dot{m}$):

$c_1 = 658.9665$ $c_2 = 21.4502$ $c_3 = -1.9168$
$c_4 = 0.26285$ $c_5 = -0.02081$ $c_6 = 0.03971$
$c_7 = 0.002069$ $c_8 = 0.0001988$ $c_9 = 0.0003108$
$c_{10} = -0.0004297$

The mass flow rate obtained from the polynomial is expressed in kg/h and is converted to kg/s within the model.

B. ADDITIONAL TABLES

The full set of simulation results is summarized in Table 9, including operating conditions and performance indicators.

Table 9: Summary of simulation results

$T_{\text{w,in}}$ (°C)	T_{evap} (°C)	T_{cond} (°C)	Q_e (kW)	P_c (kW)	EER (-)	PR (-)	η_{is} (-)
20	4.84	30.11	55.89	8.43	6.63	2.04	0.701
24	4.97	34.02	54.36	9.13	5.96	2.25	0.740
28	5.10	37.93	52.79	9.89	5.34	2.47	0.768
30	5.16	39.89	51.99	10.29	5.05	2.58	0.779
34	5.30	43.81	50.35	11.15	4.51	2.83	0.792
40	5.51	49.68	47.75	12.60	3.79	3.23	0.797

C. Code Structure and Modular Implementation

C.1 Repository Overview

The implementation is hosted on GitHub at:

https://github.com/Amirhossein-Naeejnezhad/refrigeration-cycle-model

The repository contains two branches:

Branch	Description
`main`	Original implementation — author's project only (R32, data center, 50 kW)
`universal-model`	Generalized version — supports any course project via configuration

C.2 Module Structure

The code is organized into independent modules, each responsible for a single component of the model:

File	Responsibility
`src/config.py`	All project-specific parameters (refrigerant, temperatures, compressor)
`src/compressor.py`	Compressor registry and polynomial evaluation
`src/thermodynamics.py`	CoolProp-based cycle state reconstruction
`src/heat_exchanger.py`	LMTD KA design and temperature solvers
`src/solver.py`	Iterative operating-point coupling
`src/plots_ph_ts.py`	P–h and T–s diagram generation
`src/plots_others.py`	EER, Q_e, P_c, efficiency plots
`src/utils.py`	Unit conversions, LMTD helper, numerical utilities
`src/main.py`	Orchestration: calls all modules in sequence

C.3 Universal Model Design

The `universal-model` branch extends the original implementation to support all projects in the course assignment list. The design principle is that **no computational module contains project-specific values**. All parameters are centralized in `src/config.py` and read at runtime.

Two files require editing to run a different project:

`src/config.py` **— Project parameters.** The file contains an `ACTIVE PROJECT SETTINGS` block. The current project is commented out and replaced with a new block specifying:

- Student name and application
- Refrigerant (any CoolProp-supported fluid)
- Nominal cooling capacity
- Secondary fluid type for evaporator (`EVAP_SECONDARY`: `"air"`, `"water"`, or `"brine"`) and condenser (`COND_SECONDARY`: `"air"` or `"water"`)
- Corresponding temperatures (air inlet/outlet, or brine inlet/outlet)
- Compressor datasheet values (swept volume, pressure limits, power limit)
- Nominal operating point from manufacturer software

`src/compressor.py` **— Compressor registry.** The `COMPRESSOR_REGISTRY` dictionary stores polynomial coefficient sets indexed by compressor model name. Adding a new compressor requires inserting one entry with three lists of 10 coefficients each (for Q_e, P_c, and $\dot{m}$), obtained from the Bitzer, Copeland, or Frascold selection software.

C.4 Fluid-Type Handling

A key generalization in the `universal-model` branch is the support for different secondary fluids on both sides. The `heat_exchanger.py` module determines the correct temperature pairs at runtime:

- **Air-cooled evaporator:** uses `T_AIR_IN_C` and `T_AIR_OUT_C` from config
- **Water/brine evaporator:** uses `T_BRINE_IN_C` and `T_BRINE_OUT_C` from config
- **Water-cooled condenser:** secondary inlet = heat-sink temperature; outlet = inlet + `T_WATER_RISE_K`
- **Air-cooled condenser:** secondary inlet = ambient temperature; outlet = inlet + `T_AIR_COND_RISE_K`

The `solver.py` module calls `cond_secondary_temps_at(T_heatsink)` to retrieve condenser secondary temperatures at each parametric step, with no fluid-type logic in the solver itself.

C.5 Execution

The project is run via `run_project.ipynb` in Google Colab or Jupyter. The notebook clones the `universal-model` branch directly, installs dependencies, and confirms the active project before execution:

```
git clone --branch universal-model \
    https://github.com/Amirhossein-Naeejnezhad/refrigeration-cycle-model.git
```

```
from src.main import run_project
df, df_print, df_states = run_project()
```

Results are exported automatically to a CSV file named after the active refrigerant and student: `refrigeration_results_<REF>_<LastName>.csv`